Practical No. 05

a).Develop an Android application using ListView with ArrayAdapter and implement item click handling.

Project Name: ListViewDemo

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <LinearLayout
        android:layout_width="409dp"
        android:layout_height="729dp"
        android:orientation="vertical"
        tools:layout_editor_absoluteX="1dp"
        tools:layout_editor_absoluteY="1dp">

        <TextView
            android:id="@+id/textView"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:paddingBottom="10dp"
            android:text="Programming Language"
            android:textSize="24sp"
            android:textStyle="bold" />

        <ListView
            android:id="@+id/listView"
            android:layout_width="match_parent"
            android:layout_height="match_parent" />
    </LinearLayout>
</androidx.constraintlayout.widget.ConstraintLayout>
```

Filename: MainActivity.java

```
package com.example.listviewdemo;

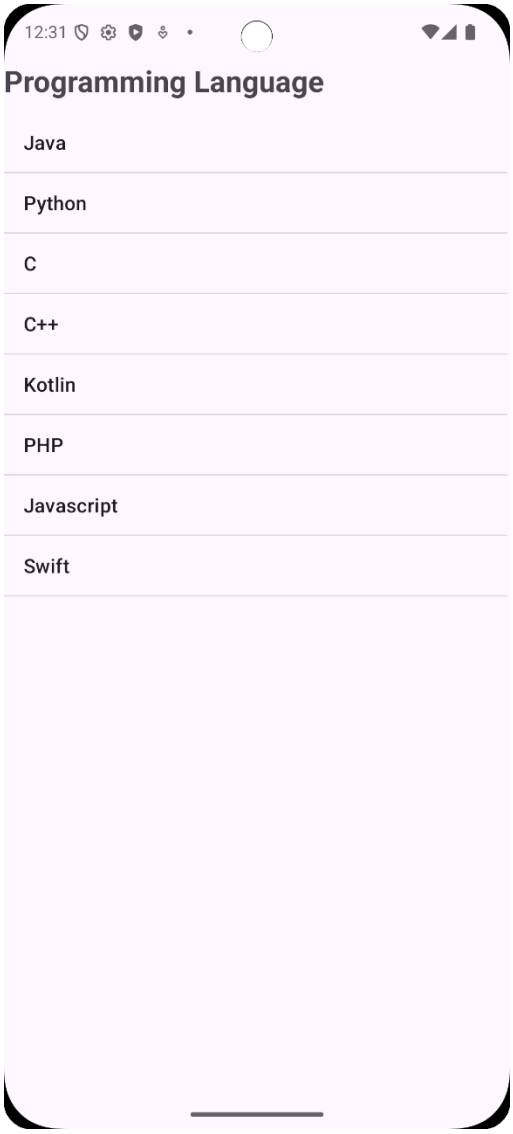
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
```

```
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    ListView listView;
    String[] languages = {
            "Java",
            "Python",
            "C",
            "C++",
            "Kotlin",
            "PHP",
            "Javascript",
            "Swift"
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        listView= findViewById(R.id.listView);
        ArrayAdapter<String> adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                languages
        );
        listView.setAdapter(adapter);
        listView.setOnItemClickListener(new AdapterView.OnItemClickListener(){
            @Override
            public void onItemClick(AdapterView<?>parent,View view,int position,
long id){
                String item = languages[position];
                Toast.makeText(
                        MainActivity.this,
                        "Selected:"+item,
                        Toast.LENGTH_SHORT
                ).show();
            }
        });
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v,
insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
    }
}
```

OUTPUT:


12:31
Programming Language
Java
Python
C
C++
Kotlin
PHP
Javascript
Swift

b).Develop an Android application using GridView with Custom Adapter and image selection handling.

Project Name: GridViewDemo

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <LinearLayout
        android:layout_width="409dp"
        android:layout_height="729dp"
        android:orientation="vertical"
        tools:layout_editor_absoluteX="1dp"
        tools:layout_editor_absoluteY="1dp">

        <TextView
            android:id="@+id/textView2"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:paddingBottom="10dp"
            android:text="Image Gallery"
            android:textSize="24sp"
            android:textStyle="bold" />

        <GridView
            android:id="@+id/gridView"
            android:layout_width="match_parent"
            android:layout_height="match_parent"
            android:numColumns="2"
            android:verticalSpacing="10dp"
            android:horizontalSpacing="10dp"
            android:stretchMode="columnWidth"/>

    </LinearLayout>
</androidx.constraintlayout.widget.ConstraintLayout>
```

Filename: grid_item.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:gravity="center"
        android:padding="10dp">
        <ImageView
            android:id="@+id/imageView"
            android:layout_width="120dp"
            android:layout_height="120dp"
            android:scaleType="centerCrop"/>
        <TextView
            android:id="@+id/textView"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:textSize="18sp"
            android:paddingTop="5dp"/>
    </LinearLayout>
```

Filename: CustomAdapter.java

</androidx.constraintlayout.widget.ConstraintLayout>

```
package com.example.grideviewdemo;
import android.content.Context;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.BaseAdapter;
import android.widget.ImageView;
import android.widget.TextView;
public class CustomAdapter extends BaseAdapter {
    Context context;
    String[] imageNames;
    int[] images;
    LayoutInflater inflater;
    public CustomAdapter(Context context, String[] imageNames, int[]images){
        this.context=context;
        this.imageNames=imageNames;
        this.images=images;
        inflater=LayoutInflater.from(context);
    }
    @Override
    public int getCount(){
```

```
        return imageNames.length;
    }
    @Override
    public Object getItem(int position) {
        return imageNames[position];
    }
    @Override
    public long getItemId(int position){
        return position;
    }
    @Override
    public View getView (int position, View convertView,ViewGroup parent){

        View view = inflater.inflate(
                R.layout.grid_item,
                parent,
                false
        );
        ImageView imageView = view.findViewById(R.id.imageView);
        TextView textView = view.findViewById(R.id.textView);
        imageView.setImageResource(images[position]);
        textView.setText(imageNames[position]);
        return view;
    }
}
```

Filename: MainActivity.java

```
package com.example.grideviewdemo;

import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.GridView;
import android.widget.Toast;

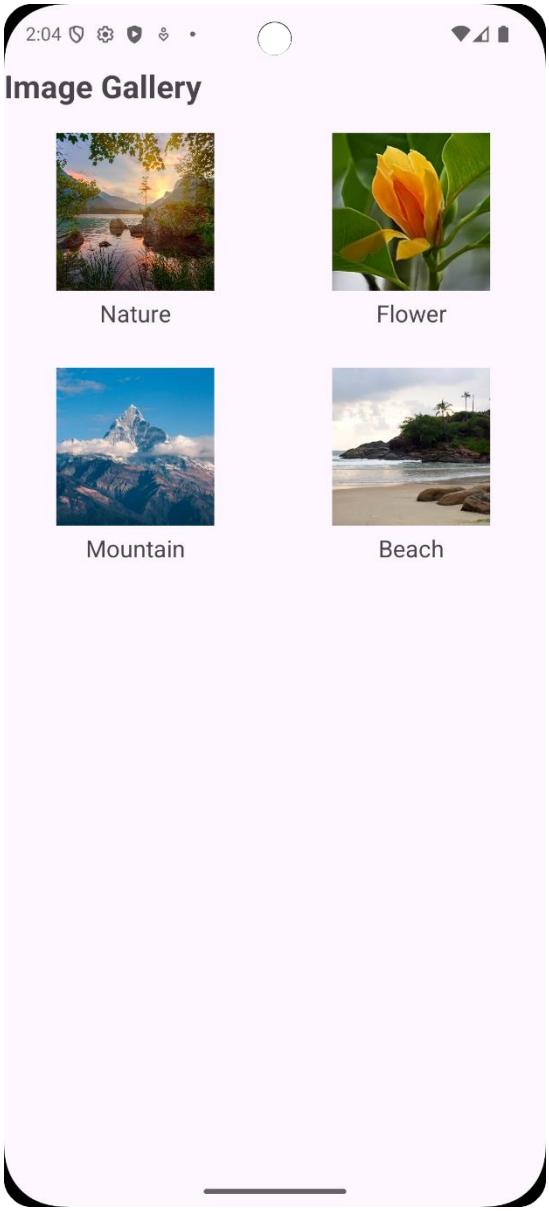
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    GridView gridView;
    String[] imageNames = {
            "Human",
            "Food",
            "Tree",
            "Animal"
    };
    int[] images = {
            R.drawable.image1,
            R.drawable.image2,
            R.drawable.image3,
```

```
            R.drawable.image4,
    };


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        gridView = findViewById(R.id.gridView);
        CustomAdapter adapter = new CustomAdapter(this, imageNames, images);
        gridView.setAdapter(adapter);
        gridView.setOnItemClickListener(
                new AdapterView.OnItemClickListener() {
                @Override
                public void onItemClick(
                        AdapterView<?>parent,
                        View view,
                        int position,
                        long id){
                    Toast.makeText(
                            MainActivity.this,
                            "Selected:"+ imageNames[position],
                            Toast.LENGTH_SHORT
                    ).show();
                }
                }
        );
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v,
insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
    }
}
```

OUTPUT:



c) Develop an Android application implementing RecyclerView as a modern alternative to ListView

Project Name: RecyclerViewDemo

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>


    <LinearLayout
        xmlns:android="http://schemas.android.com/apk/res/android"
        xmlns:app="http://schemas.android.com/apk/res-auto"
        android:id="@+id/main"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical">

        <TextView
            android:id="@+id/textView"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginVertical="50dp"
            android:text="RecyclerViewDemo"
            android:textAlignment="center"
            android:textSize="20dp"
            android:paddingBottom="10dp"
            android:textStyle="bold" />

        <androidx.recyclerview.widget.RecyclerView
            android:id="@+id/recyclerView"
            android:layout_width="match_parent"
            android:layout_height="match_parent" />
    </LinearLayout>
```

Filename: item_layout.xml

```
<?xml version="1.0" encoding="utf-8"?>
<TextView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/txtItem"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textSize="20sp"
    android:padding="15dp"/>
```

Filename: MyAdapter.java

```
package com.example.recyclerviewdemo;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

public class MyAdapter extends RecyclerView.Adapter<MyAdapter.ViewHolder> {
    String[] data;

    public MyAdapter(String[] data) {
        this.data = data;
    }

    public static class ViewHolder extends RecyclerView.ViewHolder {
        TextView txtItem;

        public ViewHolder(@NonNull View itemView) {
            super(itemView);

            txtItem = itemView.findViewById(R.id.txtItem);
        }
    }

    @NonNull
    @Override
    public ViewHolder onCreateViewHolder(
            @NonNull ViewGroup parent,
            int viewType) {
        View view = LayoutInflater.from(parent.getContext())
                .inflate(
                        R.layout.item_layout,
                        parent,
                        false
                );
        return new ViewHolder(view);
    }

    @Override
    public void onBindViewHolder(
```

```
            @NonNull ViewHolder holder,
            int position) {
        holder.txtItem.setText(data[position]);
        holder.itemView.setOnClickListener(v -> {
            Toast.makeText(
                    v.getContext(),
                    "Selected: " + data[position],
                    Toast.LENGTH_SHORT
            ).show();
        });
    }
    @Override
    public int getItemCount(){
        return data.length;
    }
}
```

Filename: MainAcitivty.java

```
package com.example.recyclerviewdemo;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import com.example.recyclerviewdemo.MyAdapter;

public class MainActivity extends AppCompatActivity {
    RecyclerView recyclerView;
    String[] languages = {
            "Java",
            "Python",
            "C",
            "C++",
            "koylin",
            "PHP",
            "JavaScript",
            "Swift",
            "Go",
            "Ruby"
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        recyclerView = findViewById(R.id.recyclerView);
        recyclerView.setLayoutManager(
                new LinearLayoutManager(this)
```

```
        );
        MyAdapter adapter = new MyAdapter(languages);
        recyclerView.setAdapter(adapter);
    }
}
```

Filename: build.gradle (:app)

```
plugins {
    alias(libs.plugins.android.application)
}

android {
    namespace 'com.example.recyclerviewdemo'
    compileSdk 36

    defaultConfig {
        applicationId "com.example.recyclerviewdemo"
        minSdk 30
        targetSdk 36
        versionCode 1
        versionName "1.0"

        testInstrumentationRunner "androidx.test.runner.AndroidJUnitRunner"
    }

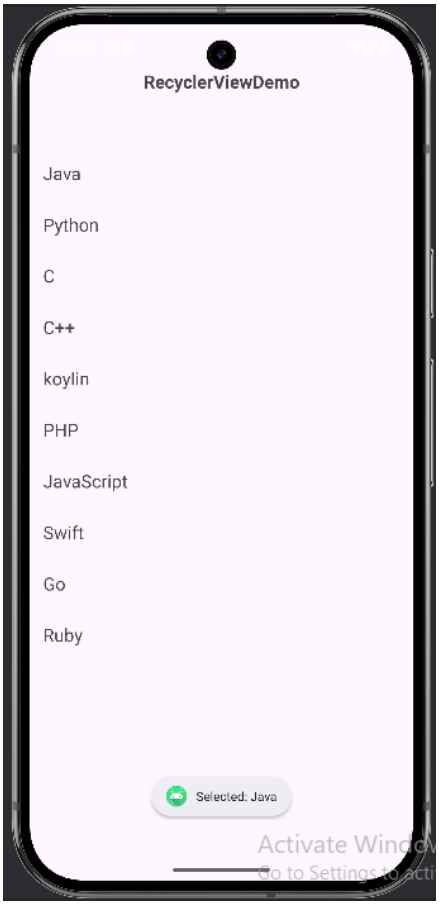
    buildTypes {
        release {
            minifyEnabled false
            proguardFiles getDefaultProguardFile('proguard-android-optimize.txt'),
'proguard-rules.pro'
        }
    }
    compileOptions {
        sourceCompatibility JavaVersion.VERSION_11
        targetCompatibility JavaVersion.VERSION_11
    }
}

dependencies {

    implementation libs.appcompat
    implementation libs.material
    implementation libs.activity
    implementation libs.constraintlayout
    testImplementation libs.junit
    androidTestImplementation libs.ext.junit
    androidTestImplementation libs.espresso.core
    implementation("androidx.recyclerview:recyclerview:1.3.2")
}
```

OUTPUT:-


RecyclerViewDemo
Java
Python
C
C++
koylin
PHP
JavaScript
Swift
Go
Ruby


RecyclerViewDemo
Java
Python
C
C++
koylin
PHP
JavaScript
Swift
Go
Ruby
Selected: Java